



Chapter 5 — Data Model & Database Design

5.1 Design Principles

- **One database, three superpowers:** PostgreSQL 16 with **PostGIS** (geospatial), **pgvector** (RAG embeddings), and **JSONB** (flexible event payloads).
- **Append-only events + mutable projections:** the `events` table is the immutable history; domain tables (`units`, `incidents`, ...) hold current state. Replay = re-project events.
- **Everything is scoped to a `run`:** a simulation run is the top-level container, making replay, comparison, and cleanup trivial.
- **Geometry column standard:** all geometry stored as `GEOMETRY(..., 4326)` (WGS84 lon/lat), converted to GeoJSON at the API layer.

5.2 Entity-Relationship Overview

```
users 1--* audit_logs
users 1--* missions (planned_by)
scenarios 1--* runs
runs 1--* units 1--* unit_positions (history)
runs 1--* incidents
runs 1--* events (append-only stream)
runs 1--* weather_cells
runs 1--* missions 1--* waypoints
runs 1--* resources (per unit/depot)
runs 1--* personnel 1--* vitals_samples
runs 1--* detections (from vision jobs)
vision_jobs 1--* detections
documents 1--* doc_chunks (RAG, run-independent)
```

`zones` (restricted/sectors, per scenario)
`forecasts` (analytics outputs, per run)

5.3 Core Tables (DDL)

```
-- Identity & access
CREATE TABLE users (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email       TEXT UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,          -- argon2id
  display_name TEXT NOT NULL,
  role        TEXT NOT NULL CHECK (role IN ('admin', 'planner', 'operator', 'viewer')),
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);

-- Scenario definitions (authored YAML) and executions (runs)
CREATE TABLE scenarios (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name        TEXT NOT NULL,
  description TEXT,
  definition  JSONB NOT NULL,          -- parsed scenario YAML
  area        GEOMETRY(POLYGON, 4326), -- operational area
  created_by  UUID REFERENCES users(id),
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE runs (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  scenario_id UUID NOT NULL REFERENCES scenarios(id),
  status      TEXT NOT NULL CHECK (status IN ('created', 'running', 'paused', 'completed', 'aborted')),
  speed       NUMERIC NOT NULL DEFAULT 1.0, -- sim-time multiplier
```

```

    sim_clock    TIMESTAMPTZ,                -- current simulated time
    started_at   TIMESTAMPTZ,
    ended_at     TIMESTAMPTZ,
    seed         BIGINT NOT NULL             -- deterministic replay
);

-- Simulated entities
CREATE TABLE units (
    id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    run_id       UUID NOT NULL REFERENCES runs(id) ON DELETE CASCADE,
    callsign     TEXT NOT NULL,              -- 'Team Bravo'
    unit_type    TEXT NOT NULL CHECK (unit_type IN
        ('ground_team', 'vehicle', 'drone', 'medical', 'depot')),
    status       TEXT NOT NULL DEFAULT 'active',
    position     GEOMETRY(POINT, 4326) NOT NULL,
    heading      NUMERIC,
    speed_kmh    NUMERIC,
    attrs        JSONB NOT NULL DEFAULT '{}' -- battery, capacity, etc.
);
CREATE INDEX idx_units_pos ON units USING GIST (position);

-- Movement history for replay (bulk-inserted per tick batch)
CREATE TABLE unit_positions (
    unit_id      UUID NOT NULL REFERENCES units(id) ON DELETE CASCADE,
    sim_time     TIMESTAMPTZ NOT NULL,
    position     GEOMETRY(POINT, 4326) NOT NULL,
    PRIMARY KEY (unit_id, sim_time)
);

CREATE TABLE incidents (
    id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```

```

    run_id      UUID NOT NULL REFERENCES runs(id) ON DELETE CASCADE,
    kind        TEXT NOT NULL,                -- 'medical','fire','flood','obstruction','observation'
    severity    TEXT NOT NULL CHECK (severity IN ('info','warning','critical')),
    status      TEXT NOT NULL DEFAULT 'open' CHECK (status IN ('open','acknowledged','resolved')),
    title       TEXT NOT NULL,
    location    GEOMETRY(POINT, 4326) NOT NULL,
    created_sim TIMESTAMPTZ NOT NULL,
    resolved_sim TIMESTAMPTZ,
    attrs       JSONB NOT NULL DEFAULT '{}';
CREATE INDEX idx_incidents_loc ON incidents USING GIST (location);

-- The append-only event stream (heart of the system)
CREATE TABLE events (
    id          BIGSERIAL PRIMARY KEY,        -- monotonically increasing: WS resume cursor
    run_id      UUID NOT NULL REFERENCES runs(id) ON DELETE CASCADE,
    sim_time    TIMESTAMPTZ NOT NULL,
    wall_time   TIMESTAMPTZ NOT NULL DEFAULT now(),
    type        TEXT NOT NULL,                -- 'UnitMove', 'IncidentCreated', ...
    source      TEXT NOT NULL,                -- 'simulation', 'vision', 'operator', 'ai'
    entity_id   UUID,
    geom        GEOMETRY(GEOMETRY, 4326),
    severity    TEXT,
    payload     JSONB NOT NULL DEFAULT '{}';
);
CREATE INDEX idx_events_run_time ON events (run_id, sim_time);
CREATE INDEX idx_events_type ON events (run_id, type);

```

```

-- Zones: restricted areas, sectors, search zones
CREATE TABLE zones (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  scenario_id UUID REFERENCES scenarios(id) ON DELETE CASCADE,
  run_id      UUID REFERENCES runs(id) ON DELETE CASCADE,
  kind        TEXT NOT NULL CHECK (kind IN ('restricted', 'sector', 'search', 'hazard')),
  name        TEXT NOT NULL,
  area        GEOMETRY(POLYGON, 4326) NOT NULL,
  attrs       JSONB NOT NULL DEFAULT '{}')
);
CREATE INDEX idx_zones_area ON zones USING GIST (area);

-- Missions & waypoints
CREATE TABLE missions (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  run_id      UUID NOT NULL REFERENCES runs(id) ON DELETE CASCADE,
  unit_id     UUID REFERENCES units(id),
  name        TEXT NOT NULL,
  kind        TEXT NOT NULL, -- 'survey', 'resupply', 'evac', 'patrol'
  status      TEXT NOT NULL DEFAULT 'draft'
  CHECK (status IN ('draft', 'planned', 'active', 'completed', 'aborted')),
  planned_by  UUID REFERENCES users(id),
  route       GEOMETRY(LINESTRING, 4326),
  estimates   JSONB NOT NULL DEFAULT '{}' -- distance_km,
  duration_min, battery_pct
);

CREATE TABLE waypoints (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  mission_id  UUID NOT NULL REFERENCES missions(id) ON DELETE CASCADE,
  seq         INT NOT NULL,
  position    GEOMETRY(POINT, 4326) NOT NULL,

```

```

    action      TEXT,                                -- 'photo','loiter','drop'
    UNIQUE (mission_id, seq)
);

-- Logistics
CREATE TABLE resources (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    run_id      UUID NOT NULL REFERENCES runs(id) ON DELETE CASCADE,
    holder_id   UUID NOT NULL,                        -- unit or depot id
    kind        TEXT NOT NULL,                        -- 'fuel','food','water','medkit','battery'
    quantity    NUMERIC NOT NULL,
    capacity    NUMERIC NOT NULL,
    updated_sim TIMESTAMPTZ NOT NULL
);

-- Personnel & synthetic vitals
CREATE TABLE personnel (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    run_id      UUID NOT NULL REFERENCES runs(id) ON DELETE CASCADE,
    unit_id     UUID REFERENCES units(id),
    name        TEXT NOT NULL,
    baseline    JSONB NOT NULL                       -- resting HR, temp, etc.
);

CREATE TABLE vitals_samples (
    person_id   UUID NOT NULL REFERENCES personnel(id) ON DELETE CASCADE,
    sim_time    TIMESTAMPTZ NOT NULL,
    heart_rate  INT, body_temp NUMERIC, spo2 NUMERIC, stress NUMERIC,
    PRIMARY KEY (person_id, sim_time)
);

```

```

-- Weather cells (moving polygons)
CREATE TABLE weather_cells (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  run_id      UUID NOT NULL REFERENCES runs(id) ON DELETE CAS
CADE,
  kind        TEXT NOT NULL,                -- 'rain','win
d','fog','heat'
  intensity   NUMERIC NOT NULL,             -- 0..1
  area        GEOMETRY(POLYGON, 4326) NOT NULL,
  velocity    JSONB NOT NULL,               -- {dx_kmh, dy_
kmh}
  sim_time    TIMESTAMPTZ NOT NULL
);

-- Vision pipeline
CREATE TABLE vision_jobs (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  run_id      UUID REFERENCES runs(id),
  video_uri   TEXT NOT NULL,               -- MinIO object
path
  status      TEXT NOT NULL DEFAULT 'queued',
  progress    NUMERIC NOT NULL DEFAULT 0,
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE detections (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id      UUID NOT NULL REFERENCES vision_jobs(id) ON DE
LETE CASCADE,
  frame_ts    NUMERIC NOT NULL,           -- seconds into
video
  class       TEXT NOT NULL,              -- 'vehicle','p
erson','smoke','fire','damage'
  confidence  NUMERIC NOT NULL,
  bbox        JSONB NOT NULL,             -- [x,y,w,h] no
rmalized
  location    GEOMETRY(POINT, 4326),      -- assigned sce

```

```

nario position
    risk_score NUMERIC,
    reviewed TEXT DEFAULT 'pending' CHECK (reviewed IN ('pe
nding','confirmed','dismissed'))
);

-- RAG knowledge base
CREATE TABLE documents (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    title TEXT NOT NULL,
    doc_type TEXT NOT NULL, -- 'sop','manua
l','map_note','incident_doc'
    source_uri TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE doc_chunks (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    doc_id UUID NOT NULL REFERENCES documents(id) ON DELET
E CASCADE,
    chunk_idx INT NOT NULL,
    content TEXT NOT NULL,
    embedding VECTOR(768) NOT NULL -- pgvector
);
CREATE INDEX idx_chunks_hnsw ON doc_chunks
    USING hnsw (embedding vector_cosine_ops);

-- Analytics outputs & audit
CREATE TABLE forecasts (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    run_id UUID NOT NULL REFERENCES runs(id) ON DELETE CA
SCADE,
    target TEXT NOT NULL, -- 'fuel_depot_
1','mission_eta_...'
    model TEXT NOT NULL, model_version TEXT NOT NULL,
    horizon JSONB NOT NULL, -- forecast poi
nts + intervals
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

```
);

CREATE TABLE audit_logs (
  id          BIGSERIAL PRIMARY KEY,
  user_id     UUID REFERENCES users(id),
  action      TEXT NOT NULL,
  target      TEXT,
  detail      JSONB,
  at          TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

5.4 Signature Geospatial Queries

```
-- 1. Units within 5 km of an incident (the Phase-1 showcase query)
SELECT u.callsign,
       ST_Distance(u.position::geography, i.location::geography) / 1000 AS km
FROM units u
JOIN incidents i ON i.id = :incident_id
WHERE u.run_id = :run_id
      AND ST_DWithin(u.position::geography, i.location::geography, 5000)
ORDER BY km;

-- 2. Does a planned route cross any restricted zone?
SELECT z.name
FROM zones z
WHERE z.kind = 'restricted'
      AND ST_Intersects(z.area, :route_linestring);

-- 3. Which units are currently inside a weather cell?
SELECT u.callsign, w.kind, w.intensity
FROM units u
JOIN weather_cells w ON ST_Contains(w.area, u.position)
WHERE u.run_id = :run_id AND w.sim_time = :now;
```

```
-- 4. Incident density per sector (drives the heatmap)
SELECT s.name, count(i.id) AS incident_count
FROM zones s
LEFT JOIN incidents i ON s.kind = 'sector' AND ST_Contains
(s.area, i.location)
WHERE s.run_id = :run_id
GROUP BY s.name;

-- 5. Replay: unit track as a GeoJSON LineString
SELECT ST_AsGeoJSON(ST_MakeLine(position ORDER BY sim_time))
FROM unit_positions WHERE unit_id = :unit_id;
```

5.5 Data Volume & Retention Strategy

Table	Growth driver	Strategy
<code>events</code>	~50/sec at full load	Partition by <code>run_id</code> ; archive completed runs to compressed dumps
<code>unit_positions</code>	positions × tick rate	Insert in per-tick batches; downsample to 1/5s for runs older than 7 days
<code>vitals_samples</code>	personnel × sample rate	Sample at 10-sim-second resolution; aggregate older data to 1-min averages
<code>detections</code>	per vision job	Keep annotated key frames only in MinIO; prune dismissed detections

5.6 Migration & Seed Strategy

- All schema changes via **Alembic** migrations — no manual DDL.
- `make seed` loads: 1 demo scenario, 3 sectors, 2 restricted zones, 8 units, 2 depots, 6 personnel per team, and the starter SOP document set for RAG.
- Deterministic seeds: `runs.seed` feeds every RNG in the engine so a rerun with the same seed reproduces the identical event stream (critical for tests and demos).